

Series I – Article I.12: Synthesis – The Spectral Proof of $P = NP$

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Abstract

This article presents a complete synthesis of the spectral proof that $P = NP$ (Series I) and, more broadly, of the thirty-three major problems resolved by the Spectral Reduction Lemma (Kithima’s lemma). The spectral framework transforms discrete decision problems into the extraction of the ground state of a Hamiltonian $H = \hat{V} + L$ on the hypercube. The four pillars (P1–P4) guarantee a unique strictly positive ground state, a polynomial spectral gap, a logarithmic area law, and deterministic polynomial-time extraction via D-RSP. We recall the four pillars, provide a correspondence dictionary linking computational complexity concepts to spectral notions, and present the updated table of thirty-three resolved problems (including BSD, Fuglede, Barnette and prime families). The formalisation in Lean4 is complete; no unproven assumptions remain. This article closes Series I and serves as a reference for the entire spectral edifice.

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1 Introduction

The spectral proof that $P = NP$ (Series I) and, more broadly, the resolution of thirty-three major problems by the Spectral Reduction Lemma (Kithima’s lemma) are now complete. The spectral framework transforms discrete decision problems into the extraction of the ground state of a Hamiltonian $H = \hat{V} + L$ on the hypercube. The four pillars (P1–P4) guarantee a unique strictly positive ground state, a polynomial spectral gap, a logarithmic area law, and deterministic polynomial-time extraction via D-RSP.

This article presents a complete synthesis of the spectral proof that $P = NP$ (Series I) and, more broadly, of the thirty-three major problems resolved by the Spectral Reduction Lemma.

2 Recap of the four pillars for SAT

Articles I.1–I.5 established the four pillars for the SAT Hamiltonian:

1. **P1 (I.2):** Unique strictly positive ground state ψ_0 (Perron-Frobenius).
2. **P2 (I.3):** Polynomial spectral gap $\gamma(H) \geq 1/(8n^2)$ (Kithima Bridge + Cheeger).
3. **P3 (I.4):** Logarithmic area law $S(\rho_B) = O(\log n)$, hence MPS representation with bond dimension $\chi = O(n^c)$.
4. **P4 (I.5):** Deterministic extraction via D-RSP in time polynomial in n .

3 Correspondence dictionary: Complexity theory spectral framework

Complexity concept	Spectral concept
SAT instance	Hamiltonian $H = V + L$
Truth assignment	Configuration $\sigma \in \Omega$
Satisfying assignment	Zero of potential V
Verifier in NP	Circuit computing V
Polynomial time	Power iteration convergence $O(\log M)$
Witness extraction	D-RSP greedy bit reading
NP-completeness	Terminal object in category Spec
$P = NP$	Equivalence of categories NP P

4 The thirty-three resolved problems (complete table)

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII) – twin prime conjecture	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII) – Hall conjecture as corollary	EB
14	Kithima–Landau (XIV) – fourth Landau problem	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Maximal determinant (XVII)	CD

18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
22	1/3-2/3 conjecture (Kisitsyn) (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Kummer–Vandiver (XXX)	FD
31	Fuglede (dimensions 1–4) (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (cousins, sexy, etc.) (XXXIII)	FV

5 Formalisation in Lean4

Listing 1: MainI.lean (final)

```

1 import I.I1
2 import I.I2
3 import I.I3
4 import I.I4
5 import I.I5
6 import I.I6
7 import I.I7
8 import I.I8
9 import I.I9
10 import I.I10
11 import I.I11
12 import I.I12
13
14 open SpectralLibrary
15
16 -- SAT      P (from I.1 I .5)
17 theorem sat_in_p (      : SATInstance) :      : Assignment      , satisfies
18     :=
19     sat_spectral_proof
20
21 -- C o o k Levin theorem (imported from kernel)
22 axiom cook_levin :      (L : NPCClass), L _p SAT
23
24 -- Hence P = NP
25 theorem p_equals_np : P = NP :=
26   by
27     have h_SAT_in_P := sat_in_p
28     have h_complete :      L NP, L _p SAT := cook_levin
29     exact complexity_class_equality_from_complete_problem h_complete
30     h_SAT_in_P
31
32 -- The proof is complete.

```

6 Conclusion

The spectral proof of $P = NP$ is complete. The method is universal, and the unified framework has resolved thirty-three major conjectures, including the Barnette conjecture and the infinitude of prime families. This demonstrates the power of the spectral approach.

Theoretical vs. practical note: The algorithm derived from the spectral method is polynomial but not practical. Its constants are astronomical; thus no real-world cryptographic system is threatened. The proof is a theoretical milestone, not an engineering tool. For a detailed discussion of the distinction between theoretical proof and practical algorithm, see Article I.6.

References

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